

CityCare: Feature Breakdown Structure (WBS)

City-Wide Hospital Emergency Resource Coordination Platform (Blood Bank MVP)

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Agile Software Project Management

1.0 User & Hospital Management

1.1 Authentication - MVP

Lets a user securely create an account and log in to the platform.

- User Accounts
- Login & Session Security

1.2 Roles & Permissions - MVP

Controls what each type of user can see and do inside the system.

- Role Management
- Access Control

1.3 Hospital Management - MVP

Lets hospitals join the network and maintain their profile, structure and capabilities.

- Hospital Registration
- Hospital Structure
- Hospital Capability Profile

2.0 Resource Management

2.1 Bed Management - MVP

Tracks bed types, real-time availability and occupancy across hospital wards.

- Bed Registry
- Bed Availability
- Capacity Monitoring

2.2 Equipment Management - MVP

Tracks critical medical equipment, its location and operational status.

- Equipment Registry
- Equipment Status

2.3 Operating Room Management - MVP

Manages operating room details and scheduling to avoid conflicts.

- Operating Room Registry
- Operating Room Scheduling

2.4 Specialist Availability - (Secondary)

Tracks specialist doctors and their on-call/availability status for emergencies.

- Doctor Profiles
- On-Call and Availability

3.0 Blood Management

3.1 Donor Management - MVP

Manages donor records, eligibility and donation scheduling.

- Donor Registration
- Donor Eligibility Screening
- Donation Appointments

3.2 Blood Inventory - MVP

Tracks blood units from collection through to expiry or use.

- Blood Collection and Testing
- Blood Unit Registration
- Expiry and Stock Control

3.3 Blood Requests & Matching - MVP

Lets hospitals request blood and finds compatible, available units to fulfil it.

- Blood Request
- Compatibility Matching
- Blood Allocation

4.0 Emergency Request Management

4.1 Emergency Case - MVP

Lets hospital staff log a new emergency case and classify its severity.

- Create Emergency Case
- Severity and Priority

4.2 Resource Requirements - MVP

Captures what resources an emergency case needs and checks what's already on hand.

- Define Requirements
- Availability Check

4.3 Request Lifecycle - MVP

Tracks an emergency request from creation through to resolution or escalation.

- Request Status Tracking
- Escalation and Closure

5.0 Matching & Allocation Engine

5.1 Candidate Search - MVP

Finds other hospitals in the network that could supply a needed resource.

- Find Candidate Hospitals
- Distance and Travel Time

5.2 Scoring & Ranking - MVP

Ranks candidate hospitals/resources so the best option is offered first.

- Scoring Model
- Ranking and Explanation
- Multi-Resource Matching

5.3 Reservation & Allocation - MVP

Holds and confirms a resource for a request so it can't be claimed twice.

- Resource Reservation
- Conflict Prevention
- Allocation Record

6.0 Transfers & Logistics

6.1 Resource Transfer - MVP

Coordinates moving a physical resource (e.g. blood units) between hospitals.

- Transfer Request
- Transfer Tracking
- Delivery Confirmation

6.2 Ambulance Management - MVP

Tracks ambulances and manages their dispatch to emergencies or transfers.

- Ambulance Registry
- Dispatch and Trip Tracking

6.3 Patient Transfer - (Secondary)

Coordinates moving a patient to a hospital that has the resources they need.

- Transfer Need Check
- Destination Selection
- Transfer Completion

7.0 Dashboards, Notifications & Reports

7.1 Notifications - MVP

Alerts the right users when something needs their attention.

- Notification Service
- Alerts

7.2 Dashboards - MVP

Gives hospital staff and city coordinators a live view of resources and emergencies.

- Hospital Dashboard
- City Command Centre

7.3 Reports - (Secondary)

Summarises historical operational data for review and planning.

- Operational Reports
- Report Export

8.0 Security & Audit

8.1 Audit Trail - MVP

Keeps a traceable record of important actions taken across the system.

- Action Logging
- Audit Search & Export